

Quantifying the Incremental Value of Retrieval-Side Reliability Signals for Selective Extractive Question Answering

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Abstract—A question-answering system that never declines will eventually present a fabricated span with the same confidence as a correct one. Training a classifier to anticipate reader errors is an established remedy, but it was formulated without a retrieval stage, and later retrieval-aware variants assume a large generative model. How much error-prediction information the retrieval stage itself contributes, once a reader-side calibrator is already fitted, has not been isolated for pipelines that cannot afford a second model pass. We measure that increment. Our system runs entirely on a CPU with no pretrained neural weights: latent-semantic and BM25 retrieval, a two-stage feature-based span reader with a learned no-answer head, sixteen reliability signals drawn from retrieval, reading and cross-passage agreement, a gradient-boosted risk estimator, and isotonic calibration. Evaluated on 3,585 questions from 36 withheld SQuAD 2.0 articles with article-level partitioning, a shuffled-label control returns an error-prediction ROC-AUC of 0.4887 inside its pre-registered 0.45–0.55 band. The combined signal set reaches an error-prediction ROC-AUC of 0.6390 (95% bootstrap interval 0.6127–0.6668) where a reader-only calibrator reaches 0.5242, and reduces the area under the risk-coverage curve by 0.0345 (paired 95% bootstrap interval for $\Delta\text{AURC} = [-0.0495, -0.0191]$, excluding zero); the ordering is unchanged at correctness cut-offs of 0.4, 0.5 and 0.6. Withholding feature families implicates retrieval most clearly (ΔAURC 0.0274) and agreement next (0.0103), while withholding all five reader signals leaves an interval spanning zero (0.0067). The added layer costs 1.96 ms of a 63.77 ms query. These findings are bounded by a deliberately simple reader: base accuracy is 0.1144 and three of seven pre-registered gates failed. We claim the measurement, not the mechanism.

Index Terms—selective prediction, question answering, abstention, confidence calibration, uncertainty estimation, information retrieval, resource-constrained inference

I. INTRODUCTION

Silence is not a failure mode that document question-answering systems support well. Given a query whose answer is absent from the collection, a retrieval-based reader still emits a span, still attaches it to a genuine passage, and still formats it exactly as it would format a correct response. Nothing in the output separates the two cases, so a cautious user must

check every answer, which erases the reason for deploying the system.

Deciding whether to publish an extracted span is therefore a separate problem from extracting it. The reader’s own probability is a poor instrument for that decision. It is normalised over candidate spans inside a passage that was handed to the reader, so it can express which span is locally best but not whether the passage deserved to be retrieved. Two situations are consequently invisible to it: the collection holds no answer, and the retriever ranked the relevant passage out of reach. Together these describe 29.5% and 6.1% of our evaluation set.

Statistics produced during retrieval do see that region, and they cost nothing extra. Similarity values, the gap between competing passages and the shape of the score distribution are already computed before the reader begins. Whether independently retrieved passages converge on the same string is equally cheap.

None of this is a new observation. Selective prediction rests on established theory [8], [9]; error-predicting calibrators for question answering were introduced by Kamath, Jia and Liang [1]; reading calibration at scale was examined by Dhuliawala *et al.* [3]; and the sufficiency of retrieved context has been studied from several directions [4]–[7]. **We make no claim to have introduced retrieval-aware abstention.**

The quantity that remains unmeasured is the increment. Holding fixed a reader-side calibrator of the type established in [1], how much further error-prediction information do retrieval and cross-passage signals supply under matched partitions and interval estimation? And does that increment persist once the pipeline is restricted to a CPU with no pretrained weights, the regime where the low cost of these signals matters most? This paper reports both measurements.

Contributions.

- 1) A controlled procedure for separating the reliability information carried by retrieval, reader and agreement signals, built on article-level partitions, a shuffled-label control and article-clustered bootstrap intervals.
- 2) A sixteen-signal reliability representation derived solely from quantities the pipeline has already computed,

H. Kamat, H. Ranjan and M. Kumar performed the implementation and experiments. D. Tyagi supervised the work as project guide. S. Kesh served as project coordinator. Institutional e-mail addresses and ORCID identifiers are omitted pending confirmation.

adding no inference pass.

- 3) A comparison against five alternatives: unconditional answering, two single-signal thresholds, the reader’s own no-answer head, and a reader-only calibrator reproducing [1] in our pipeline.
- 4) Family-withholding ablation with a permutation cross-check, a transfer evaluation on a disjoint collection, an outcome taxonomy, and per-stage cost measurement.
- 5) Evidence that the combined representation sharpens error discrimination and lowers selective risk against the reader-only calibrator in this CPU-only setting, reported alongside the pre-registered criteria it did not meet.

II. RELATED WORK

Selective prediction. Pairing a predictor with a gate that may withhold a decision was formalised by El-Yaniv and Wiener [8] and carried to deep models by Geifman and El-Yaniv [9]. Performance is summarised by the risk–coverage curve, whose area (AURC) we adopt as the primary quantity. Smaller values indicate better selective behaviour.

Calibration. Guo *et al.* [10] documented that modern classifiers report unreliable confidence and compared post-hoc corrections. Isotonic regression as a calibration map originates with Zadrozny and Elkan [11]. We use it and report expected calibration error (ECE) on either side of the correction.

Error prediction in reading systems. Kamath, Jia and Liang [1] demonstrated that a classifier trained to anticipate reader mistakes outperforms thresholding the reader’s softmax, most visibly under domain shift. Because their reader receives a designated passage, retrieval-derived signals fall outside their formulation. We reproduce their approach over our reader’s outputs as baseline B5 and treat it as the target to beat rather than a convenient foil. Dhuliawala *et al.* [3] reached a compatible conclusion at larger scale. The unanswerable questions of SQuAD 2.0 [2] are what make abstention measurable at all.

Sufficiency of retrieved context. Joren *et al.* [4] define *sufficient context* and observe that capable models frequently answer when the retrieved material cannot support an answer, proposing sufficiency-guided selective generation. Abdumalikov *et al.* [5] report that rejection models trained against randomly paired passages transfer poorly to distractors sharing vocabulary with the query, accuracy collapsing from 98% to 1%, and repair this using SQuAD 2.0 unanswerable pairs. Muhamed *et al.* [6] perturb grounding material systematically and find refusal accuracy under 50% on multi-document inputs, further separating refusal into detection and categorisation abilities. Lewis *et al.* [7] show that retrieval scores alone permit selective answering along a tunable accuracy–coverage curve.

Positioning. Collectively this work establishes the theory of selective prediction, the advantage of trained error predictors over raw reader confidence, and the diagnostic value of context sufficiency. It does not isolate the *marginal* contribution of retrieval and agreement signals *given* a reader-side calibrator in a pipeline barred from a second large-model pass. The nearest studies [4], [6], [7] operate with large generative models

or large neural retrievers. *We claim the measurement, not the mechanism.*

III. PROBLEM FORMULATION AND SCOPE

Question. Do retrieval-side and cross-evidence reliability signals supply incremental error-prediction value beyond a reader-only calibrator in a CPU-only extractive question-answering pipeline?

Secondary questions concern whether reader error is predictable at all here, whether the combination surpasses the reader’s own no-answer head, whether the resulting estimate is calibrated, which family is responsible, whether the estimator transfers to an unseen collection, and what it costs.

Formulation. For a query q , a collection C , a retrieved subset $R \subseteq C$ and an extracted span $\hat{a}$, we require $g(q, R, \hat{a}) \rightarrow [0, 1]$ approximating $P(\hat{a} \text{ is wrong})$, together with a cut-off τ such that responding only when $g < \tau$ realises a chosen operating point.

Correctness. An answerable query counts as handled correctly when the returned span attains SQuAD F1 ≥ 0.5 against any reference; an unanswerable query counts as handled correctly only when nothing is returned. The prediction target is the complement. SQuAD normalisation precedes scoring. The framing assumes an incorrect answer is more costly than silence, which is a property of the deployment rather than something we establish.

IV. DATA AND PROTOCOL

A. Answerability once retrieval is introduced

The SQuAD 2.0 label is defined against a single paragraph: a query marked unanswerable was unanswerable *from the paragraph shown to the annotator*. Retrieval dissolves that guarantee. Once other paragraphs enter the collection an “unanswerable” query may become answerable, and the label quietly ceases to describe what is being scored.

We sized the exposure before fixing the design. Each unanswerable SQuAD 2.0 item carries a `plausible_answer`: the most persuasive incorrect span from its own paragraph. Occurrence of that string elsewhere in the collection is a concrete path to an invalid label. Counting over 40 randomly drawn training articles, and restricting attention to substantive strings (≥ 2 tokens, not purely numeric, ≥ 6 characters, $n = 3,049$), **14.89%** of such strings reappear when an article’s own sibling paragraphs are pooled into the collection, against **2.39%** when they are excluded — a sixfold reduction.

Roughly one substantive unanswerable label in seven would therefore be compromised by pooling, which exceeds any effect we intend to detect. We adopt the **sibling-masked** arrangement: a query may reach its own source paragraph and any paragraph from a *different* article, never another paragraph of its own article.

Answerable labels survive because the source paragraph always remains reachable; failure to rank it inside the top- k is then a real retrieval error of the kind under study. Unanswerable labels survive because the only same-topic paragraph in scope is the one against which the annotator judged the

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query
↓ RETRIEVAL TF-IDF → SVD(256) → inner product
  BM25Okapi; fuse  $\alpha=0.5$ ; sibling mask
↓ top- $k$  paragraphs + scores
  READER 1. keep 3 best-matching sentences
          2.  $n$ -grams  $\leq 8$ , drop stopwords/punct edges
          3. GBM span scorer (20 span features)
          4. GBM no-answer head → P(no answer)
↓ span, span prob., no-answer score, entropy
  RELIABILITY LAYER (measured)
    retrieval(7) + reader(5) + agreement(4)
    GBM → isotonic → P(error)
↓
risk >  $\tau$  ? yes → abstain + reason
           no → answer + paragraph + risk

```

Fig. 1. Pipeline. Every block was implemented and exercised in the reported experiments; the cut-off τ is fixed on the calibration partition only.

query unanswerable. Masking consults the article identifier alone and is applied uniformly to both classes; conditioning it on the label would import the target into the collection. The construction is the familiar distractor arrangement from multi-document QA, used here to protect labels rather than to raise difficulty.

Paragraphs enter whole. A median SQuAD paragraph is 107 tokens, inside the reader’s span window, so we do not subdivide; doing so would cut reference spans at boundaries and add a confound unrelated to the question.

B. Collections and partitions

Collection A supplies the main experiment: 120 articles sampled from SQuAD 2.0 train, 5,637 paragraphs, 11,921 queries, 31.17% unanswerable. Collection B is the complete SQuAD 2.0 development set, reserved for transfer: 35 articles, 1,204 paragraphs, 3,500 queries, 50.46% unanswerable. Their article titles do not intersect.

Partitions are cut **by article**. SQuAD supplies a median of 279 queries per article, many revisiting the same facts, so dividing at the query level scatters near-duplicates across both sides and inflates every estimate. Three partitions do not suffice, because two models are fitted: a reliability estimator trained on articles the reader has already absorbed would observe an error distribution far more favourable than the one it meets at test time. Collection A is therefore split 40/20/10/30 by article into `reader_train` (48 articles), `rel_train` (24 articles, 2,400 queries), `rel_calib` (12 articles, 1,200 queries) and `test` (36 articles, 3,585 queries). Disjointness is asserted by a unit test that fails the build.

C. Pre-registration and interval estimation

Seven numeric criteria were recorded before execution and none was revised afterwards. Every run uses seed 20260814; hyperparameters remain at library defaults and were never tuned against the test partition. Intervals come from a cluster bootstrap over **articles** with 1,000 resamples, since queries within an article are dependent and resampling queries would understate the spread. Two-method comparisons use a paired bootstrap over identical article resamples.

V. SYSTEM

A. Retrieval

Paragraphs become TF-IDF vectors compressed to 256 dimensions by truncated SVD — latent semantic analysis in the sense of Deerwester *et al.* [12] — and are searched exactly by inner product. A BM25Okapi index [13] operates alongside. Both score vectors are min-max scaled across the candidate set and averaged with equal weight. The compressed representation absorbs paraphrase; the lexical index restores exact names, figures and identifiers that a 256-dimensional projection dilutes, and yields the query-paragraph overlap statistic that later proves the most load-bearing reliability signal.

B. Reader

The first stage orders a paragraph’s sentences by overlap with the query’s content words and retains three. Measured on training articles, 68.2% of reference answers occur in the single best sentence and 83.3% within the best three, so the candidate space contracts sharply at modest cost. The second stage enumerates token n -grams up to length eight inside those sentences, removes any span whose first or last token is a stopword or punctuation — a filter that preserves 90.4% of reference answers while discarding 64% of candidates — and scores what remains with a gradient-boosted classifier over twenty span descriptors covering capitalisation, numerals, inverse document frequency, proximity to query terms, sentence overlap, surrounding context and interactions with the interrogative type. A second gradient-boosted model, fitted on the same articles against the `is_impossible` flag, emits $P(\text{this paragraph holds no answer})$. The reader thus presents the same interface as a transformer trained on SQuAD 2.0 — a best span, a span score, an explicit no-answer score — without pretrained weights.

Constraint. A DistilBERT reader and MiniLM embeddings were specified originally. Neither could be retrieved: every route to the model repository was blocked in our build environment. Rather than fabricate numbers or abandon the study, we substituted established methods needing no pretrained weights while preserving the interface the question depends on. **No transformer, learned embedding or pretrained language model appears anywhere in the evaluated system.** Section X quantifies what this costs.

C. Reliability signals and the gate

Sixteen signals in three families (Table I), each read from quantities the pipeline has already produced. A gradient-boosted classifier maps them to a risk score; isotonic regression [11] fitted on the calibration partition converts that score into an estimated error probability; a cut-off is then chosen on the same partition.

Two points require care. **Isotonic regression aligns the score with observed error frequency; it certifies nothing.** The system estimates risk and applies a cut-off, and Section VII-D records an operating point it failed to realise. Second, no signal triggers an additional model pass, which is what holds the layer at 1.96 ms instead of doubling inference, and

TABLE I
THE SIXTEEN RELIABILITY SIGNALS.

Family	Signals
Retrieval (7)	best fused score; separation between the two best; mean over top- k ; entropy of the score distribution over a wider list; count exceeding a similarity floor; query-paragraph lexical overlap; query length
Reader (5)	span probability; no-answer score; their difference; span length; entropy of the span score distribution
Agreement (4)	paragraphs among the top- k yielding a non-empty span; edit similarity between the two leading spans; share of top- k spans equal to the modal span; whether the selected span recurs in the second paragraph

TABLE II
DECISION PROCEDURES COMPARED.

ID	Procedure	Signal
B1	Answer unconditionally	none (floor)
B2	Span-probability cut-off	reader softmax
B3	Retrieval-score cut-off	best fused score
B4	Reader no-answer head	the reader's own trained head
B5	Reader-only calibrator, after [1]	the 5 reader signals
P1	Combined calibrator (evaluated)	all 16 signals

why richer signals available in the literature were excluded deliberately.

VI. BASELINES AND ABLATION DESIGN

The six procedures in Table II consume identical pipeline output over identical partitions; only the signal driving the decision differs. **The comparison that answers the question is P1 against B5**, not P1 against B1.

The ablation withholds one family at a time and additionally fits each family alone — seven configurations settled in advance. Permutation importance over the full model supplies an independent attribution. A shuffled-label control runs first; unless it returns to chance, no downstream figure is reported.

VII. RESULTS

A. Control

Permuting the error labels and refitting yields an error-prediction ROC-AUC of **0.4887** on the test partition, inside the pre-registered band $[0.45, 0.55]$. Criterion G0 is met; no leakage is detectable through the signals, the partitions or the calibration step.

B. Main comparison

Table III gives all six procedures. P1 leads throughout, and its interval is disjoint from those of B2, B4 and B5.

Two entries deserve emphasis. B2 — a cut-off on reader confidence, the reflexive choice in practice — produces an interval straddling 0.5 and is thus not separable from chance

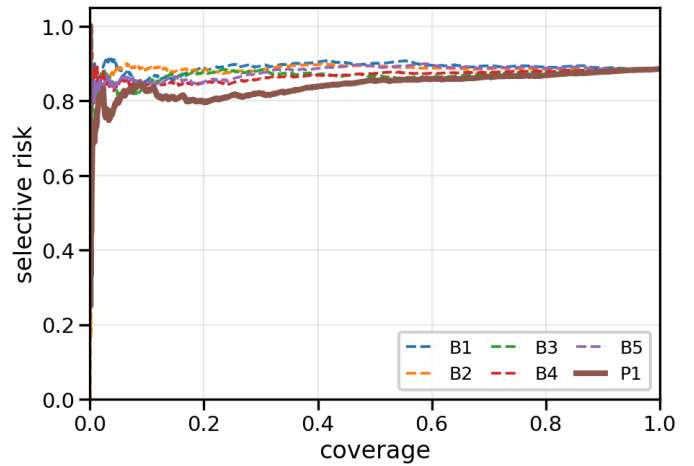


Fig. 2. Risk-coverage behaviour of all six procedures on the test partition. Lower curves indicate lower selective risk; P1 generally lies below the base-lines across the coverage range.

here. And the two retrieval-aware single signals, B3 (0.5634) and B4 (0.5526), both exceed the reader-only calibrator B5 (0.5242), despite B5 combining five signals through a fitted model.

C. The increment

Paired bootstrap over articles, identical resamples for both procedures:

- $\Delta\text{AURC} (\text{P1} - \text{B5}) = -0.0345$, 95% interval $[-0.0495, -0.0191]$, excluding zero. P1 *reduces* AURC relative to B5. Criterion G3 is met.
- $\Delta\text{AURC} (\text{P1} - \text{B4}) = -0.0293$, 95% interval $[-0.0442, -0.0150]$, excluding zero. Criterion G4 is met.

These are paired-bootstrap intervals excluding zero. No null-hypothesis test was performed and we do not describe the outcome as statistically significant.

The 0.5 correctness cut-off is a choice, so the comparison was repeated at 0.4 and 0.6. P1 reaches an error-prediction ROC-AUC of 0.6431 and 0.6352 against 0.5132 and 0.5354 for B5, with ΔAURC of -0.0438 $[-0.0633, -0.0263]$ and -0.0294 $[-0.0445, -0.0169]$. Direction and interval behaviour are unaffected by the cut-off.

D. Calibration and an operating point that was not reached

Isotonic regression lowers expected calibration error from **0.0895 to 0.0120**, inside the 0.10 criterion, and Fig. 3 shows the corresponding movement toward the diagonal.

The intended operating point was not achieved. A cut-off fixed on the calibration partition for a target accuracy of 80% attained 60% accuracy there at 0.42% coverage, and on the test partition produced **28% accuracy at 0.7% coverage** — short of target by 0.52. **Criterion G5 is not met.** The reason is visible in Table III: base accuracy is 0.1144, and while the risk estimate orders queries better than chance it cannot isolate a subset that is 80% accurate. Every procedure, P1

TABLE III
SELECTIVE PREDICTION OVER 3,585 TEST QUERIES FROM 36 WITHHELD ARTICLES. INTERVALS ARE 95% CLUSTER BOOTSTRAP OVER ARTICLES.
LOWER AURC IS BETTER. SELACC@*c* DENOTES SELECTIVE ACCURACY AT COVERAGE *c*.

Procedure	Error ROC-AUC	95% interval	AURC	95% interval	SelAcc@0.5	SelAcc@0.7
B1 answer unconditionally	0.5000	—	0.8890	0.8455–0.9162	0.0988	0.1064
B2 span probability	0.4920	0.4544–0.5304	0.8823	0.8498–0.9084	0.1066	0.1143
B3 retrieval score	0.5634	0.5353–0.5925	0.8686	0.8362–0.8962	0.1328	0.1263
B4 reader no-answer head	0.5526	0.5216–0.5823	0.8683	0.8365–0.8982	0.1267	0.1235
B5 reader-only calibrator [1]	0.5242	0.5080–0.5428	0.8785	0.8412–0.9019	0.1071	0.1112
P1 combined (evaluated)	0.6390	0.6127–0.6668	0.8423	0.8029–0.8731	0.1462	0.1355

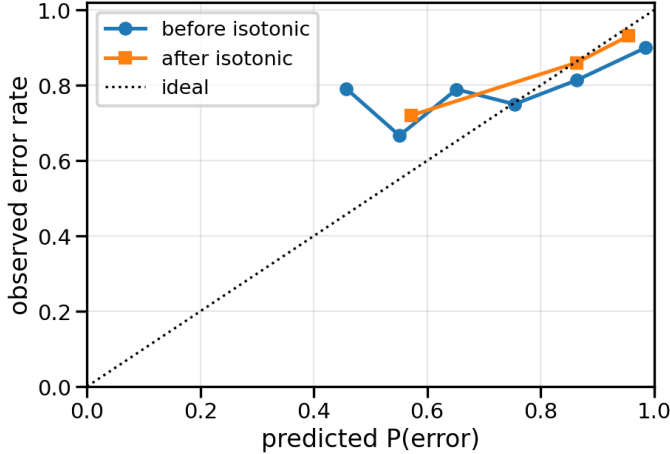


Fig. 3. Reliability of the risk estimate before and after isotonic correction. Expected calibration error falls from 0.0895 to 0.0120.

included, reports zero coverage at that target. Here the target is unreachable rather than merely missed, and **the system must not be presented as delivering any guaranteed accuracy level.**

E. Which family is responsible

Table IV reports the ablation. Withholding retrieval costs roughly four times what withholding the reader family costs, and withholding all five reader signals yields an interval that spans zero.

Permutation importance over the full model points the same way. The most load-bearing signals are query–paragraph lexical overlap (retrieval, Δ AURC 0.0121 when permuted), recurrence of the span in the second paragraph (agreement, 0.0107), span length (reader, 0.0103), span probability (reader, 0.0078) and best fused score (retrieval, 0.0066). The reader’s dedicated no-answer signal registers -0.0017 , indistinguishable from noise.

A caveat applies to any ablation over correlated inputs: because the families encode overlapping information, withholding one bounds its contribution from below rather than partitioning credit exactly. That two attribution methods agree strengthens the reading without making it exact.

VIII. ERROR ANALYSIS AND TRANSFER

Span selection, not retrieval, dominates the failures (Table V). The source paragraph enters the top five for 89.1% of queries, after which the reader selects a non-overlapping span in more than half of all cases; only 6.1% fail because retrieval missed. **The reader, not the retriever, is the bottleneck.**

Outcomes vary with interrogative type. Constrained answer forms fare best — *when* 18.8% ($n=256$), *how many* 18.7% ($n=257$), *who* 16.1% ($n=366$) — because the span descriptors encode numerals and capitalisation. Open forms fare worst: *what* 8.9% ($n=1,713$), *why* 1.7% ($n=59$).

Three cases drawn verbatim from the test partition illustrate the behaviour. Asked how many passengers passed through Brasília’s airport in 2007, the reader returned **11,119** where the reference is **11,119,872**; the truncated numeral scores below cut-off and the risk estimate assigned it the highest value in the partition, which is the correct decision. Asked for the fourth stomach compartment in ruminants, it returned **abomasum** against **The abomasum** — correct after normalisation, and assigned the lowest risk. For the unanswerable query about who appointed the governor of the Federal District before 1988, it returned **Brazilian Federal Senate** at low risk: an entity of the expected type sitting in a high-overlap sentence defeats every signal we compute, and this is precisely the failure the agreement family was meant to catch and does not.

Across collections the P1 point estimate shows no large degradation (0.6494 against 0.6390 within-collection). We did not run a paired test across collections and therefore do not assert that degradation is absent, only that the point estimate does not display it. B5 does not transfer comparably (0.5177). P1 fitted on A also exceeds P1 fitted on B itself (0.6494 against 0.5680), which we read as a sample-size effect — A supplies 2,400 reliability-training queries against B’s 700 — rather than any benefit of transfer. The setting shifts collection and reader instance together and does not separate the two.

IX. COST

The reliability layer takes 1.96 ms, near 3% of a 63.77 ms query, so criterion G6 is met with two orders of magnitude to spare (Table VII). Retrieval dominates. Peak resident memory reached 210.3 MB, index construction over 5,637 paragraphs took 4.64 s, and reader fitting 29.2 s. Stored artefacts total 58.4 MB. Nothing contacts the network at query time. These

TABLE IV
FAMILY-WITHHOLDING ABLATION. Δ AURC IS MEASURED AGAINST THE FULL SIXTEEN-SIGNAL MODEL; POSITIVE VALUES INDICATE THAT REMOVAL DEGRADED PERFORMANCE.

Configuration	Signals	Error AUC	AURC	Δ AURC vs full	95% interval	Excludes zero
Full (R+D+A)	16	0.6390	0.8423	—	—	—
– Retrieval	9	0.5410	0.8698	+0.0274	0.0149–0.0411	yes
– Agreement	12	0.6021	0.8513	+0.0103	0.0043–0.0173	yes
– Reader	11	0.6202	0.8467	+0.0067	–0.0044–0.0189	no
Retrieval only	7	0.5662	0.8640	+0.0222	0.0088–0.0352	yes
Agreement only	4	0.5680	0.8694	+0.0283	0.0138–0.0420	yes
Reader only	5	0.5242	0.8785	+0.0345	0.0191–0.0495	yes

TABLE V
OUTCOME TAXONOMY OVER ALL 3,585 TEST QUERIES.

Outcome	Count	Share
Wrong span (source paragraph retrieved)	1,844	51.4 %
Unanswerable query answered anyway	1,056	29.5 %
Correct	410	11.4 %
Retrieval failure (source not in top- k)	219	6.1 %
Partial overlap below cut-off	56	1.6 %

figures describe *this* system; they are small chiefly because no transformer is present, and should not be read as a statement about a transformer pipeline.

Of seven pre-registered criteria, four were met (G0 control 0.4887; G3 Δ AURC -0.0345 ; G4 Δ AURC -0.0293 ; G6 layer 1.96ms) and three were not: base reader accuracy 0.1144 against a 0.45 requirement (G1), a best error-prediction ROC-AUC of 0.6390 against a 0.70 requirement (G2), and the operating point of Section VII-D (G5).

X. LIMITATIONS

The reader bounds every result. Absent pretrained weights, the reader is a classical feature-based span scorer. Base accuracy is 0.1144 and mean answer F1 is 0.145, so G1 fails by a wide margin and every finding is conditional on this reader. A no-answer head trained on 130,319 queries rather than our 4,736 might carry considerably more independent signal, which would raise the reader family’s relative contribution and narrow our conclusion to weak-reader regimes. Repeating the measurement with a transformer reader is the most consequential outstanding step, and the implementation is backend-agnostic to permit it.

Error prediction is modest. G2 fails; the best value is 0.6390. The estimate orders queries better than chance and is useful for ranking, but is not strong discrimination.

No accuracy guarantee. G5 fails and every procedure sits at zero coverage for the 80% target. Calibration succeeded in the score-to-frequency sense and failed in the operating-point sense.

Residual label risk. After sibling masking, 2.39% of substantive unanswerable queries retain a plausible-answer string somewhere in the collection. The string test is a proxy: pres-

ence does not establish answerability, and paraphrase could make a query answerable with no string match.

Adversarial construction. SQuAD 2.0’s unanswerable queries were written to look plausible, and may be systematically easier or harder to detect than those arising naturally. This dataset cannot resolve the question.

Collection, language, granularity, scale. English Wikipedia only, with no technical, multilingual or specialist evaluation. Paragraph-level retrieval sidesteps a segmentation confound but leaves the results inapplicable to systems that must divide long documents. A single CPU with an exact index over 5,637 paragraphs gives no evidence about million-paragraph collections under approximate search.

Transfer and attribution. One transfer setting, moving collection and reader instance together, without a paired test across collections. Withholding a family of correlated signals bounds its contribution from below rather than partitioning credit.

No deployment evidence. No user study, no field trial, no production data. The system is a research prototype operating over a fixed collection; it is not claimed to be production-ready, and no comparison to any commercial system is made or implied.

XI. CONCLUSION

We measured how much error-prediction information retrieval-side and cross-evidence signals add beyond a reader-only calibrator in a CPU-only extractive question-answering pipeline. Over 3,585 withheld SQuAD 2.0 queries from 36 unseen articles, with article-level partitions and a passing shuffled-label control, the combined representation raised error-prediction ROC-AUC from 0.5242 to 0.6390 and lowered AURC by 0.0345 with a paired-bootstrap interval excluding zero. The ordering persisted at three correctness cut-offs and under transfer to a disjoint collection, and the layer cost 1.96ms of a 63.77ms query. Ablation and permutation importance both attribute the contribution primarily to retrieval signals and secondarily to agreement, with the reader family’s independent contribution spanning zero.

Absolute quality is weak. Base reader accuracy is 0.1144, no procedure reached the pre-registered error-prediction target, the intended 80% operating point is unreachable at any useful coverage, and three of seven criteria failed. We report

TABLE VI
RELIABILITY ESTIMATOR FITTED ON COLLECTION A AND EVALUATED ON COLLECTION B, WHOSE ARTICLES ARE DISJOINT. ECE: EXPECTED CALIBRATION ERROR.

Procedure	Setting	n	Error AUC	AURC	ECE	Base error rate
P1	A $\rightarrow$ A within-collection	3,585	0.6390	0.8423	0.0120	0.8856
P1	A $\rightarrow$ B transfer	1,000	0.6494	0.8612	0.0143	0.9000
P1	B $\rightarrow$ B within-collection	1,000	0.5680	0.8843	0.0479	0.9000
B5	A $\rightarrow$ A within-collection	3,585	0.5242	0.8785	0.0134	0.8856
B5	A $\rightarrow$ B transfer	1,000	0.5177	0.9003	0.0029	0.9000
B5	B $\rightarrow$ B within-collection	1,000	0.5862	0.8773	0.0527	0.9000

TABLE VII
PER-QUERY COST, SINGLE CPU, $n = 7,185$ QUERIES; GATE TIMED OVER 100 REPETITIONS.

Stage	Median (ms)	p95 (ms)
Retrieval	43.54	60.99
Reader	19.83	23.80
Reliability signals	0.22	0.32
Risk model + gate	1.74	2.06
End-to-end query	63.77	83.03
Reliability layer	1.96	2.37

the incremental finding because it was measured, held under several analysis choices and is inexpensive to adopt; we report the failures because a stronger reader could change them, and because a reader unaware of them would be misled.

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